

OKN1 – OKN1 TASK 1: USING AI TO RESOLVE CHALLENGES

ADVANCED AI FOR COMPUTER SCIENTISTS – D804

PRFA – OKN1

Preparation

Task Overview

Submissions

Evaluation Report

COMPETENCIES

4142.1.1 : Prepares Data

The learner prepares data for use in AI model training.

4142.1.2 : Synthesizes Principles of AI and Machine Learning

The learner synthesizes principles of artificial intelligence and machine learning to design systems that solve real-world problems.

4142.1.3 : Applies Probabilistic Reasoning

The learner applies probabilistic reasoning and AI techniques for enhanced problem-solving capabilities.

4142.1.4 : Evaluates AI Systems

The learner critically evaluates AI systems to improve performance, efficiency, and sustainability.

INTRODUCTION

This performance assessment will evaluate your ability to integrate the advanced artificial intelligence techniques with the data engineering skills from this course. You will develop a series of mini projects that demonstrate some of the techniques covered in this course. The assessment will focus on your proficiency in evaluating, designing, implementing, and validating AI systems that leverage complex data structures and algorithms to solve challenges.

REQUIREMENTS

Your submission must represent your original work and understanding of the course material. Most performance assessment submissions are automatically scanned through the WGU similarity checker. Students are strongly encouraged to wait for the similarity report to generate after uploading their work and then review it to ensure Academic Authenticity guidelines are met before submitting the file for evaluation. See [Understanding Similarity Reports](#) for more information.

Grammarly Note:

Professional Communication will be automatically assessed through Grammarly for Education in most performance assessments before a student submits work for evaluation. Students are strongly encouraged



review the Grammarly for Education feedback prior to submitting work for evaluation, as the overall submission will not pass without this aspect passing. See [Use Grammarly for Education Effectively](#) for more information.

Microsoft Files Note:

Write your paper in Microsoft Word (.doc or .docx) unless another Microsoft product, or pdf, is specified in the task directions. Tasks may not be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc. All supporting documentation, such as screenshots and proof of experience, should be collected in a pdf file and submitted separately from the main file. For more information, please see [Computer System and Technology Requirements](#).

A. Create your subgroup and project in GitLab using the provided "GitLab" and the "GitLab How-To" web links by doing the following:

- Clone the project to the IDE.
- Commit with a message and push when you complete each requirement listed in parts B-I.

Note: You may commit and push whenever you want to back up your changes, even if a requirement is not yet complete.

- Submit a copy of the GitLab repository URL in the "Comments to Evaluator" section when you submit this assessment.
- Submit a copy of the repository branch history retrieved from your repository, which must include the commit messages and dates.

Note: Wait until you have completed all the following prompts before you create your copy of the repository branch history.

B. Choose one dataset from an open source for each of your models.

C. Identify the goal of the first model, including the name of the model using the format "D804_PA_Model_Title" where "Title" is the name you have chosen the model.

D. Identify the dataset to be used for the first model.

1. Describe the dataset and its features.
2. Explain why you chose this dataset.
3. Describe the technique(s) used to prepare the data for modeling.

E. Develop the first model in the Cloud Academy Lab Environment using any of the advanced methods from the course, identifying which method you chose.

1. Explain how the model was developed.
2. Explain how the model was trained to accomplish its task.
3. Explain how the results were validated for the model's performance.
4. Explain how the model could be deployed to end users.
5. Document the code for submission to GitLab.

- F. Identify the goal of the second model, including the name of the model using the format “D804_PA_Model_Title” where “Title” is the name you have chosen for the model.
- G. Identify the dataset to be used for the second model.
1. Describe the dataset and its features.
 2. Explain why you chose this dataset.
 3. Describe the technique(s) used to prepare the data for modeling.
- H. Develop the second model in the Cloud Academy Lab Environment using any of the probabilistic methods from the course, identifying which method you chose.
1. Explain how the model was developed.
 2. Explain how the model was trained to accomplish its task.
 3. Explain how the results were validated for the model's performance.
 4. Explain how the model could be deployed to end users.
 5. Document the code for submission to GitLab.
- I. Identify the goal of the optimization for each model, including the name of the project using the format “D804_PA_Optimization_Title” where “Title” is the name you have chosen.
- J. Perform the following actions on **both** of the models:
- Optimize the efficiency and performance of the model.
 - Identify benchmarks for evaluation.
 - Evaluate the model based on the identified benchmarks.
- K. Acknowledge sources, using APA-formatted in-text citations and references, for content that is quoted, paraphrased, or summarized.
- L. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ()

File size limit: 200 MB

File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A:GITLAB

NOT EVIDENT

The submission does not establish a project and subgroup in GitLab.

APPROACHING COMPETENCE

The submission does not accurately establish a project and subgroup in GitLab.

COMPETENT

The submission accurately establishes a project and subgroup in GitLab.

B:DATASET CHOICE

NOT EVIDENT

The submission does not include 2 datasets.

APPROACHING COMPETENCE

The submission includes only 1 dataset.

COMPETENT

The submission includes 2 datasets.

C:MODEL GOAL

NOT EVIDENT

The submission does not identify the goal of the first model and the file format is incorrect.

APPROACHING COMPETENCE

The submission incorrectly identifies the goal of the first model, or the file format is incorrect.

COMPETENT

The submission correctly identifies the goal of the first model and the file format is correct.

D:DATASET IDENTIFICATION

NOT EVIDENT

The dataset for the first model is not identified.

APPROACHING COMPETENCE

The dataset for the first model is incorrectly identified.

COMPETENT

The dataset for the first model is correctly identified.

D1:DATASET DESCRIPTION

NOT EVIDENT

The features of the first dataset are not described.

APPROACHING COMPETENCE

The described features of the first dataset are incorrect.

COMPETENT

The described features of the first dataset are correct.

D2:DATASET REASONING

NOT EVIDENT**APPROACHING COMPETENCE****COMPETENT**

The submission does not include a reason for choosing the dataset.

The submission includes an irrelevant reason for choosing the dataset.

The submission includes a relevant reason for choosing the dataset.

D3:DATA PREPARATION**NOT EVIDENT**

The technique(s) used for data preparation are not described.

APPROACHING COMPETENCE

The technique(s) used for data preparation are described incorrectly.

COMPETENT

The described technique(s) used for data preparation are correct.

E1:MODEL DEVELOPMENT METHOD**NOT EVIDENT**

The submission does not explain how the model was developed, or which method was chosen.

APPROACHING COMPETENCE

The submission explains how the model was developed but omits which method was chosen.

COMPETENT

The submission explains how the model was developed, and which method was chosen.

E2:MODEL TRAINING**NOT EVIDENT**

The submission does not explain how the model was trained.

APPROACHING COMPETENCE

The submission incorrectly explains how the model was trained.

COMPETENT

The submission correctly explains how the model was trained.

E3:RESULTS VALIDATION**NOT EVIDENT**

The submission does not explain how the results were validated for the model's performance.

APPROACHING COMPETENCE

The submission incorrectly explains how the results were validated for the model's performance.

COMPETENT

The submission correctly explains how the results were validated for the model's performance.

E4:MODEL DEPLOYMENT

NOT EVIDENT

The submission does not explain how the model could be deployed to end users.

APPROACHING COMPETENCE

The submission incorrectly explains how the model could be deployed to end users.

COMPETENT

The submission correctly explains how the model could be deployed to end users.

E5:CODE DOCUMENTATION**NOT EVIDENT**

The submission does not include code documentation.

APPROACHING COMPETENCE

The submission includes incomplete code documentation.

COMPETENT

The submission includes complete code documentation.

F:MODEL GOAL**NOT EVIDENT**

The submission does not identify the goal of the second model and the file format is incorrect.

APPROACHING COMPETENCE

The submission incorrectly identifies the goal of the second model, or the file format is incorrect.

COMPETENT

The submission correctly identifies the goal of the second model and the file format is correct.

G:DATASET IDENTIFICATION**NOT EVIDENT**

The dataset for the second model is not identified.

APPROACHING COMPETENCE

The dataset for the second model is incorrectly identified.

COMPETENT

The dataset for the second model is correctly identified.

G1:DATASET DESCRIPTION**NOT EVIDENT**

The features of the second dataset are not described.

APPROACHING COMPETENCE

The described features of the second dataset are incorrect.

COMPETENT

The described features of the second dataset are correct.

G2:DATASET REASONING**NOT EVIDENT**

The submission does not include a reason for choosing the dataset.

APPROACHING COMPETENCE

The submission includes an irrelevant reason for choosing the dataset.

COMPETENT

The submission includes a relevant reason for choosing the dataset.

G3:DATA PREPARATION**NOT EVIDENT**

The technique(s) used for data preparation are not described.

APPROACHING COMPETENCE

The described technique(s) used for data preparation are incorrect.

COMPETENT

The technique(s) used for data preparation are described correctly.

H1:MODEL DEVELOPMENT METHOD**NOT EVIDENT**

The submission does not explain how the model was developed, or which method was chosen.

APPROACHING COMPETENCE

The submission explains how the model was developed but omits which method was chosen.

COMPETENT

The submission explains how the model was developed, and which method was chosen.

H2:MODEL TRAINING**NOT EVIDENT**

The submission does not explain how the model was trained.

APPROACHING COMPETENCE

The submission incorrectly explains how the model was trained.

COMPETENT

The submission correctly explains how the model was trained.

H3:RESULTS VALIDATION**NOT EVIDENT**

The submission does not explain how the results were validated for the model's performance.

APPROACHING COMPETENCE

The submission incorrectly explains how the results were vali-

COMPETENT

The submission correctly explains how the results were vali-

dated for the model's performance.

dated for the model's performance.

H4:MODEL DEPLOYMENT

NOT EVIDENT

The submission does not explain how the model would be deployed to end users.

APPROACHING COMPETENCE

The submission incorrectly explains how the model would be deployed to end users.

COMPETENT

The submission correctly explains how the model would be deployed to end users.

H5:CODE DOCUMENTATION

NOT EVIDENT

The submission does not include code documentation.

APPROACHING COMPETENCE

The submission includes incomplete code documentation.

COMPETENT

The submission includes complete code documentation.

I:GOALS OF OPTIMIZATION

NOT EVIDENT

The goals for the optimization of *both* models and the titles for the models are not provided.

APPROACHING COMPETENCE

The goal of only 1 model is provided, or the goals for either model are incomplete or either project title is incorrect.

COMPETENT

The goals of *both* models are provided with the correct project names.

J:MODEL IMPROVEMENTS

NOT EVIDENT

None of the required actions are performed on either model.

APPROACHING COMPETENCE

All the required actions were performed on only 1 of the models or only some of the required actions were performed on *both* models.

COMPETENT

All the required actions are performed on *both* models.

K:APA SOURCES

NOT EVIDENT

The submission does not include in-text citations and references according to APA style for content that is quoted, paraphrased, or summarized.

APPROACHING COMPETENCE

The submission includes in-text citations and references for content that is quoted, paraphrased, or summarized but does not demonstrate a consistent application of APA style.

COMPETENT

The submission includes in-text citations and references for content that is quoted, paraphrased, or summarized and demonstrates a consistent application of APA style.

L:PROFESSIONAL COMMUNICATION**NOT EVIDENT**

This submission includes pervasive errors in professional communication related to grammar, sentence fluency, contextual spelling, or punctuation, negatively impacting the professional quality and clarity of the writing. Specific errors have been identified by Grammarly for Education under the Correctness category.

APPROACHING COMPETENCE

This submission includes substantial errors in professional communication related to grammar, sentence fluency, contextual spelling, and/or punctuation. Specific errors have been identified by Grammarly for Education under the Correctness category.

COMPETENT

This submission includes satisfactory use of grammar, sentence fluency, contextual spelling, and punctuation, which promote accurate interpretation and understanding.

WEB LINKS

[GitLab How-To](#)

[Gitlab](#)

[Cloud Academy Lab Environment](#)